Perform the following operations using Python on the Air quality data sets a. Data cleaning b. Data integration c. Data transformation d. Error correcting e. Data model building

In [1]:
```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score,confusion_matrix
from sklearn.linear_model import LogisticRegression
import seaborn as sns
import matplotlib.pyplot as plt
```

In [2]:
```
df = pd.read_csv("data_airquality.csv")
df.head()
```

```
---------------------------------------------------------------------------
UnicodeDecodeError                        Traceback (most recent call last)
Cell In[2], line 1
----> 1 df = pd.read_csv(                    )
      2 df.head()

File ~\anaconda3\Lib\site-packages\pandas\io\parsers\readers.py:1026, in read_csv
(filepath_or_buffer, sep, delimiter, header, names, index_col, usecols, dtype, en
gine, converters, true_values, false_values, skipinitialspace, skiprows, skipfoot
er, nrows, na_values, keep_default_na, na_filter, verbose, skip_blank_lines, pars
e_dates, infer_datetime_format, keep_date_col, date_parser, date_format, dayfirs
t, cache_dates, iterator, chunksize, compression, thousands, decimal, linetermina
tor, quotechar, quoting, doublequote, escapechar, comment, encoding, encoding_err
ors, dialect, on_bad_lines, delim_whitespace, low_memory, memory_map, float_preci
sion, storage_options, dtype_backend)
   1013 kwds_defaults = _refine_defaults_read(
   1014     dialect,
   1015     delimiter,
   (...)   1022     dtype_backend=dtype_backend,
   1023 )
   1024 kwds.update(kwds_defaults)
-> 1026 return _read(filepath_or_buffer, kwds)

File ~\anaconda3\Lib\site-packages\pandas\io\parsers\readers.py:626, in _read(fil
epath_or_buffer, kwds)
    623     return parser
    625 with parser:
--> 626     return parser.read(nrows)

File ~\anaconda3\Lib\site-packages\pandas\io\parsers\readers.py:1923, in TextFile
Reader.read(self, nrows)
   1916 nrows = validate_integer("nrows", nrows)
   1917 try:
   1918     # error: "ParserBase" has no attribute "read"
   1919     (
   1920         index,
   1921         columns,
   1922         col_dict,
-> 1923     ) = self._engine.read(  # type: ignore[attr-defined]
   1924         nrows
   1925     )
   1926 except Exception:
   1927     self.close()

File ~\anaconda3\Lib\site-packages\pandas\io\parsers\c_parser_wrapper.py:234, in
CParserWrapper.read(self, nrows)
    232 try:
    233     if self.low_memory:
--> 234         chunks = self._reader.read_low_memory(nrows)
    235         # destructive to chunks
    236         data = _concatenate_chunks(chunks)

File pandas/_libs/parsers.pyx:838, in pandas._libs.parsers.TextReader.read_low_me
mory()

File pandas/_libs/parsers.pyx:905, in pandas._libs.parsers.TextReader._read_rows
()

File pandas/_libs/parsers.pyx:874, in pandas._libs.parsers.TextReader._tokenize_r
ows()
```

```
File pandas/_libs/parsers.pyx:891, in pandas._libs.parsers.TextReader._check_toke
nize_status()

File pandas/_libs/parsers.pyx:2053, in pandas._libs.parsers.raise_parser_error()

File ~\anaconda3\Lib\codecs.py:325, in BufferedIncrementalDecoder.decode(self, in
put, final)
    322 def decode(self, input, final=False):
    323     # decode input (taking the buffer into account)
    324     data = self.buffer + input
--> 325     (result, consumed) = self._buffer_decode(data, self.errors, final)
    326     # keep undecoded input until the next call
    327     self.buffer = data[consumed:]

UnicodeDecodeError: 'utf-8' codec can't decode byte 0x96 in position 154449: inva
lid start byte
```

In [3]:
```
!pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in c:\users\rohit\anaconda3\lib\site-
packages (1.7.2)
Requirement already satisfied: numpy>=1.22.0 in c:\users\rohit\anaconda3\lib\site
-packages (from scikit-learn) (2.3.5)
Requirement already satisfied: scipy>=1.8.0 in c:\users\rohit\anaconda3\lib\site-
packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in c:\users\rohit\anaconda3\lib\site
-packages (from scikit-learn) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\rohit\anaconda3\l
ib\site-packages (from scikit-learn) (3.5.0)
```

In [4]:
```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score,confusion_matrix
from sklearn.linear_model import LogisticRegression
import seaborn as sns
import matplotlib.pyplot as plt
```

In [5]:
```
df = pd.read_csv(
    "data_airquality.csv",
    encoding='latin1',
    low_memory=False
)
df.head()
```

Out[5]:

	stn_code	sampling_date	state	location	agency	type	so2	no2	rspm
0	150	February - M021990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	4.8	17.4	NaN
1	151	February - M021990	Andhra Pradesh	Hyderabad	NaN	Industrial Area	3.1	7.0	NaN
2	152	February - M021990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	6.2	28.5	NaN
3	150	March - M031990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	6.3	14.7	NaN
4	151	March - M031990	Andhra Pradesh	Hyderabad	NaN	Industrial Area	4.7	7.5	NaN

In [6]:

```
#Data cleaning
df=df.drop_duplicates()
df.describe()
```

Out[6]:

	so2	no2	rspm	spm	pm2_5
count	400436.000000	418846.000000	395033.000000	198160.000000	9314.000000
mean	10.830468	25.823296	108.871711	220.774796	40.791467
std	11.168994	18.497878	74.898637	151.374791	30.832525
min	0.000000	0.000000	0.000000	0.000000	3.000000
25%	5.000000	14.000000	56.000000	111.000000	24.000000
50%	8.000000	22.000000	90.000000	187.000000	32.000000
75%	13.700000	32.200000	142.000000	296.000000	46.000000
max	909.000000	876.000000	6307.033333	3380.000000	504.000000

In [7]:

```
df.isna().sum()
```

Out[7]:
```
stn_code                       144077
sampling_date                       3
state                               0
location                            3
agency                         149466
type                             5357
so2                             34632
no2                             16222
rspm                            40035
spm                            236908
location_monitoring_station     27303
pm2_5                          425754
date                                7
dtype: int64
```

In [8]:
```
percent_missing = df.isnull().sum() * 100 / len(df)
percent_missing.sort_values(ascending=False)
```

Out[8]:
```
pm2_5                          97.859185
spm                            54.453097
agency                         34.354630
stn_code                       33.115973
rspm                            9.202010
so2                             7.960135
location_monitoring_station     6.275571
no2                             3.728613
type                            1.231302
date                            0.001609
sampling_date                   0.000690
location                        0.000690
state                           0.000000
dtype: float64
```

In [9]:
```
df = df.drop(columns=[
    'stn_code',
    'agency',
    'sampling_date',
    'location_monitoring_station',
    'pm2_5'
], errors='ignore')

df.head()
```

Out[9]:

	state	location	type	so2	no2	rspm	spm	date
0	Andhra Pradesh	Hyderabad	Residential, Rural and other Areas	4.8	17.4	NaN	NaN	2/1/1990
1	Andhra	Hyderabad	Industrial Area	3.1	7.0	NaN	NaN	2/1/1990
2	Andhra Pradesh	Hyderabad	Residential, Rural and other Areas	6.2	28.5	NaN	NaN	2/1/1990
3	Andhra	Hyderabad	Residential, Rural and	6.3	14.7	NaN	NaN	3/1/1990
4	Andhra Pradesh	Hyderabad	Industrial Area	4.7	7.5	NaN	NaN	3/1/1990

In [10]:
```
col_var = ['state', 'location', 'type','date']
col_num = ['so2','no2','rspm','spm']
for col in df.columns:
    if df[col].dtype == 'object' or df[col].dtype == 'string':
        df[col] = df[col].fillna(df[col].mode()[0])
    else:
        df[col] = df[col].fillna(df[col].mean())
```

In [11]:
```
df.isna().sum()
```

Out[11]:
```
state       0
location    0
type        0
so2         0
no2         0
rspm        0
spm         0
date        0
dtype: int64
```

In [12]:
```
#Data integration
subSet1 = df[['state', 'type']]
subSet2 = df[['state','location']]
subSet1.head()
```

Out[12]:

	state	type
0	Andhra Pradesh	Residential, Rural and other Areas
1	Andhra Pradesh	Industrial Area
2	Andhra Pradesh	Residential, Rural and other Areas
3	Andhra Pradesh	Residential, Rural and other Areas
4	Andhra Pradesh	Industrial Area

In [13]:
```
subSet2.head()
```

Out[13]:

	state	location
0	Andhra Pradesh	Hyderabad
1	Andhra Pradesh	Hyderabad
2	Andhra Pradesh	Hyderabad
3	Andhra Pradesh	Hyderabad
4	Andhra Pradesh	Hyderabad

In [14]:
```
concatenated_df = pd.concat([subSet1, subSet2], axis=1)
concatenated_df
```

Out[14]:

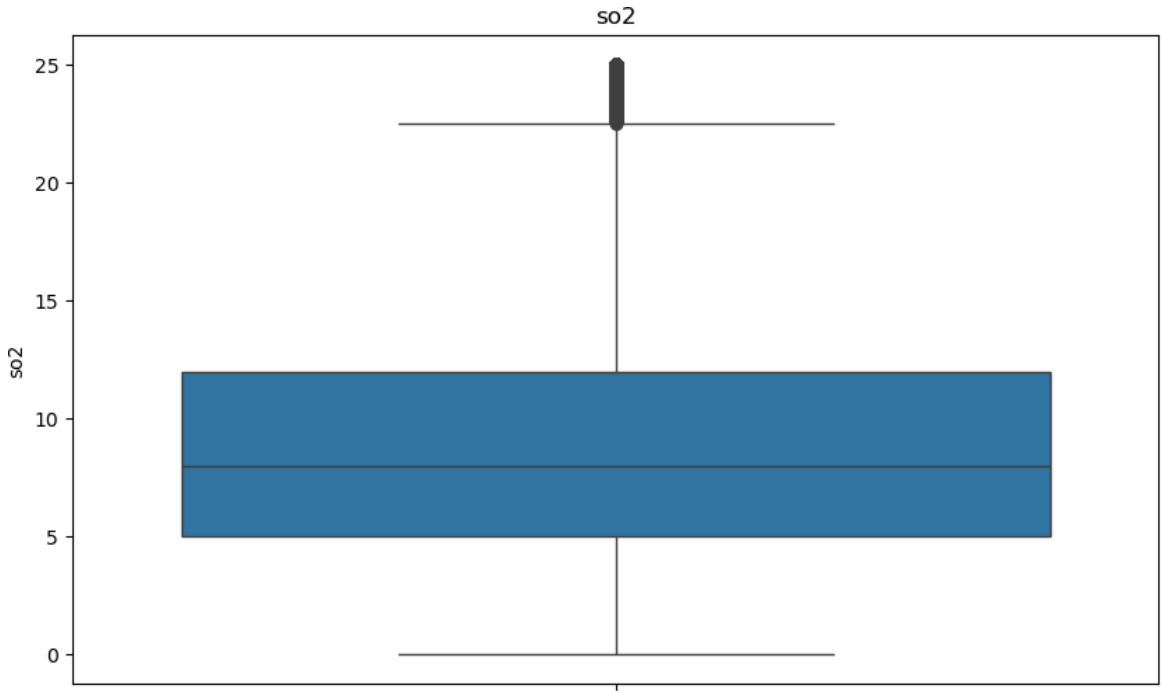
	state	type	state	location
0	Andhra Pradesh	Residential, Rural and other Areas	Andhra Pradesh	Hyderabad
1	Andhra Pradesh	Industrial Area	Andhra Pradesh	Hyderabad
2	Andhra Pradesh	Residential, Rural and other Areas	Andhra Pradesh	Hyderabad
3	Andhra Pradesh	Residential, Rural and other Areas	Andhra Pradesh	Hyderabad
4	Andhra Pradesh	Industrial Area	Andhra Pradesh	Hyderabad
...	...	...	...	...
435737	West Bengal	RIRUO	West Bengal	ULUBERIA
435738	West Bengal	RIRUO	West Bengal	ULUBERIA
435739	andaman-and-nicobar-islands	Residential, Rural and other Areas	andaman-and-nicobar-islands	Guwahati
435740	Lakshadweep	Residential, Rural and other Areas	Lakshadweep	Guwahati
435741	Tripura	Residential, Rural and other Areas	Tripura	Guwahati

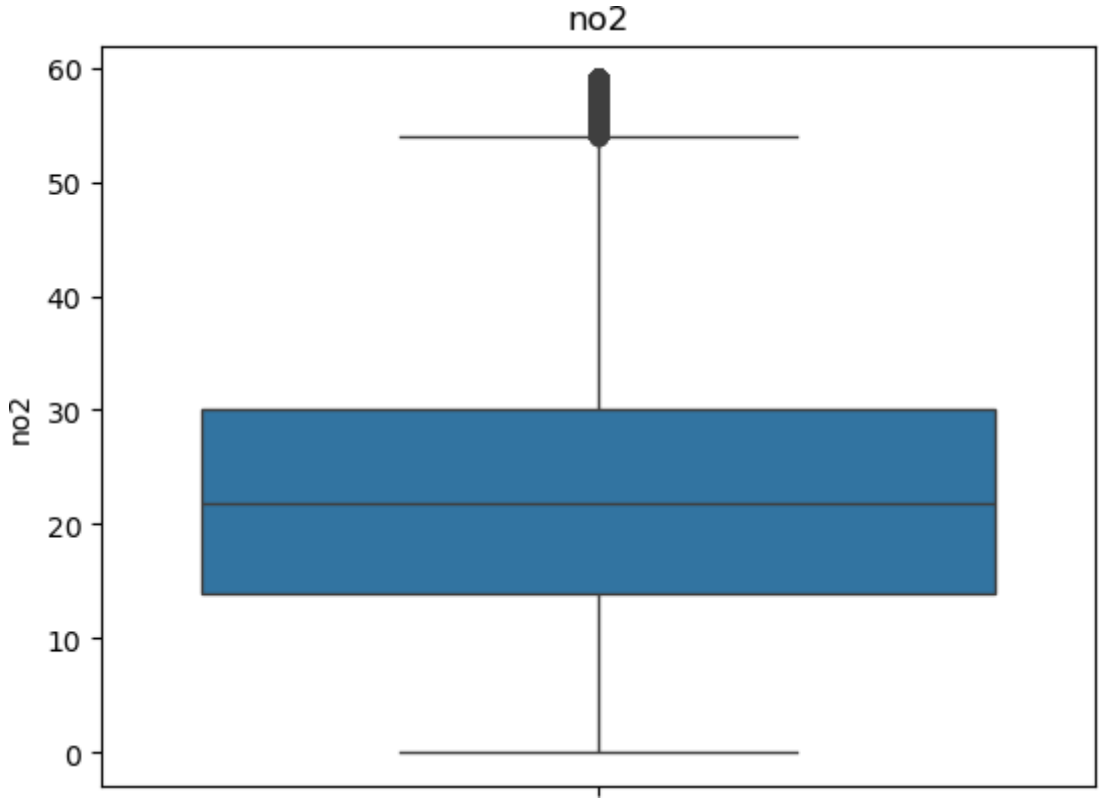
435068 rows × 4 columns

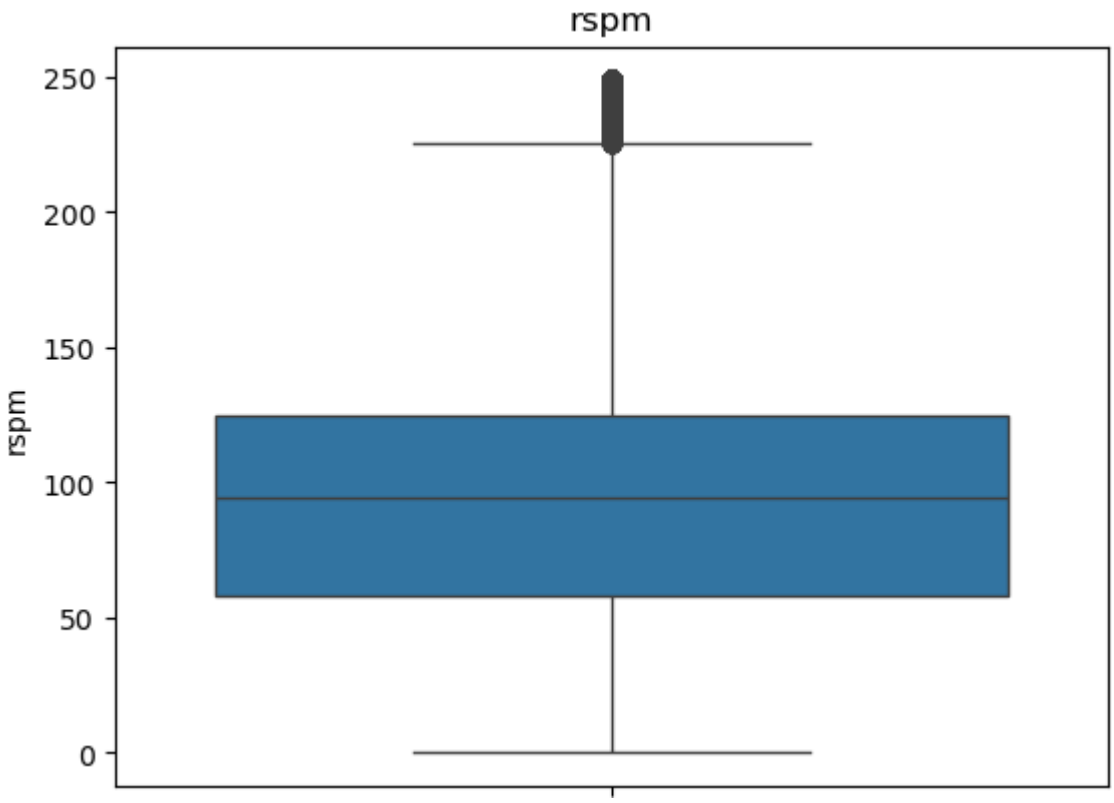
In [15]:
```
def remove_outliers(column):
    Q1 = column.quantile(0.25)
    Q3 = column.quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    return column.where((column >= lower) & (column <= upper))
```

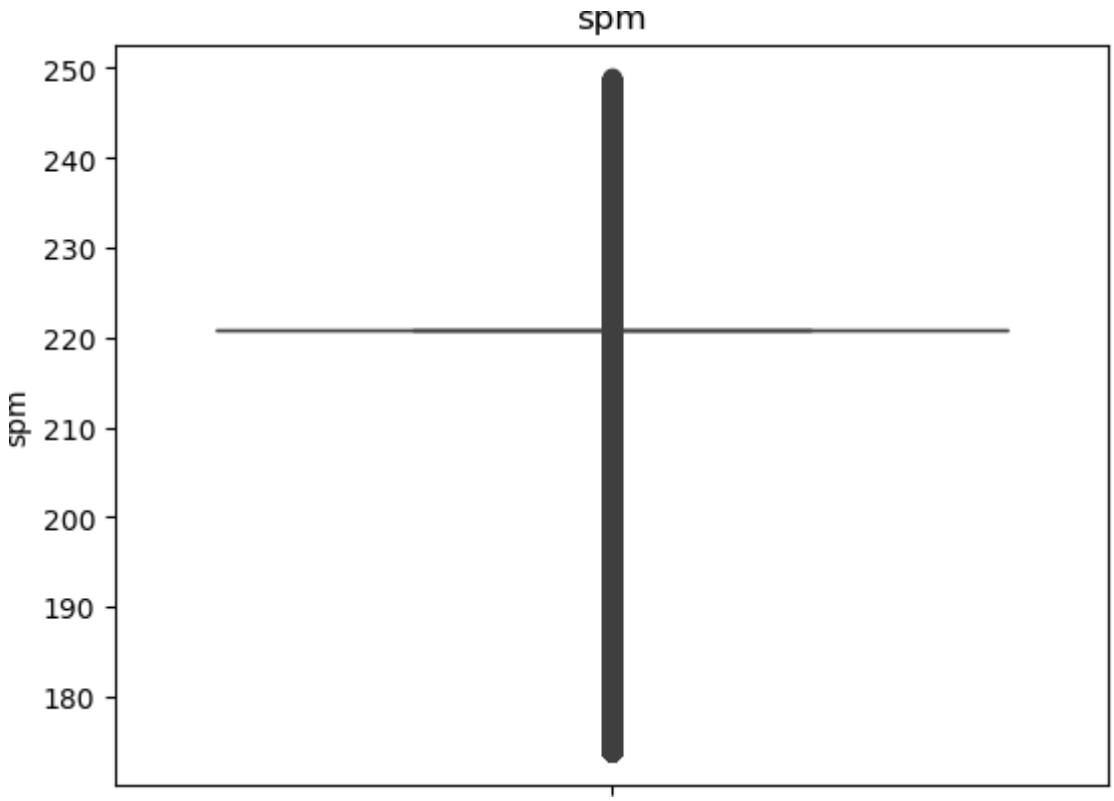
In [16]:
```
#Remove outliers
col_name = ['so2', 'no2', 'rspm', 'spm']
for col in col_name:
    df[col] = remove_outliers(df[col])
```

In [17]:
```
plt.figure(figsize=(10, 6))
# Adjust the figure size if needed
for col in col_name:
    sns.boxplot(data=df[col])
    plt.title(col)
    plt.show()
```


so2
so2
25
20
15
10
5
0


no2
no2
60
50
40
30
20
10
0


rspm
250
200
150
100
50
0
rspm


spm
250
240
230
220
210
200
190
180
spm

In [18]:
```
#Data Transform
from sklearn.preprocessing import LabelEncoder
col_label= ['state','location','type']
# Initialize LabelEncoder
encoder = LabelEncoder()
# Iterate over columns
for col in df.columns:
    # Fit and transform the column
    df[col] = encoder.fit_transform(df[col])
df
```

Out[18]:

	state	location	type	so2	no2	rspm	spm	date
0	0	114	6	446	1489	2030	464	1693
1	0	114	1	197	250	2030	464	1693
2	0	114	6	790	3096	2030	464	1693
3	0	114	6	823	1144	2030	464	2093
4	0	114	1	427	301	2030	464	2093
...	...	...	...	...	...	...	...	...
435737	35	282	3	2888	5307	2534	464	1505
435738	35	282	3	2809	5113	3098	464	1570
435739	36	100	6	1638	2696	2030	464	2256
435740	17	100	6	1638	2696	2030	464	2256
435741	31	100	6	1638	2696	2030	464	2256

435068 rows × 8 columns

In [19]:
```
# splitting data using train test split
df = df.sample(1000, random_state=0)
x  =  df[['so2','no2','spm','rspm']]
y = df['location']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=0)
x_train.shape,x_test.shape,y_train.shape,y_test.shape
```

Out[19]:
```
((800, 4), (200, 4), (800,), (200,))
```

In [20]:
```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
x_train_scaled = scaler.fit_transform(x_train)
x_test_scaled = scaler.transform(x_test)
```

In [21]:
```
model = LogisticRegression(max_iter=100)
model.fit(x_train_scaled, y_train)
y_pred = model.predict(x_test_scaled)
from sklearn.metrics import accuracy_score
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.075
```

```
C:\Users\Rohit\anaconda3\Lib\site-packages\sklearn\metrics\_classification.py:98:
UserWarning: The number of unique classes is greater than 50% of the number of sa
mples. `y` could represent a regression problem, not a classification problem.
  type_true = type_of_target(y_true, input_name="y_true")
```

In [ ]: